

Linear Algebra: Matrix Transformations, Eigenvalues & Diagonalization

Subject: MATH101 | Unit 2: Matrix Theory

1. Executive Overview & Syllabus Scope

Matrix determinants, rank, Gaussian elimination, characteristic equations, Cayley-Hamilton theorem, Eigenvalues and Eigenvectors

2. Key Architectural Invariants & Core Topics

â€¢ Lecture Modules & Key Topics

- Unit 2: Engineering Mathematics I - Matrix Theory & Linear Algebra
- 1. Systems of Linear Equations: Row Echelon Form, Rank of a Matrix, and Rouché-Capelli Theorem
- 2. Characteristic Equation: $\det(A - \lambda I) = 0$ and computation of Eigenvalues & Eigenvectors
- 3. Cayley-Hamilton Theorem: Verification and computation of inverse matrix A^{-1} and higher powers
- 4. Diagonalization: Modal Matrix P, Spectral decomposition, and Orthogonal similarity transformations

3. University Exam Tips & High-Yield Checklist

- * Practice numerical problems and time complexity derivations under timed conditions.
- * Draw neat labeled block diagrams for full credit in subjective university evaluations.
- * Use AI_Book Quiz & Summary features to test active recall before mid-terms and finals.